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SERVICE VALVE FOR AIR CONDITIONER, WHICH CAN PREVENT SPINDLE WEAR

Abstract

A service valve for use in an air conditioner may include: a body; a fluid passage formed to penetrate the body; a spindle receiving part formed to communicate with the fluid passage; a spindle screw-connected to the spindle receiving part and configured to open and block the fluid passage; a guard hole formed at a certain depth on an upper surface of the spindle; and a wrench guard disposed in the guard hole.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application is a continuation application of International Application No. PCT/KR2023/019679 designating the United States, filed on Dec. 1, 2023, in the Korean Intellectual Property Receiving Office and claiming priority to Korean Patent Application No. 10-2022-0184898, filed on Dec. 26, 2022, the disclosures of which are all hereby incorporated by reference herein in their entireties.

BACKGROUND

Technical Field

[0002] Certain example embodiments may relate to an air conditioner, and for example, to a service valve to be used in an air conditioner.

Background Art

[0003] Generally, an air conditioner may include an indoor unit and an outdoor unit.

[0004] The indoor unit may include an indoor fan and an evaporator. The outdoor unit may include a compressor, a condenser, an expansion valve, and an outdoor fan.

[0005] The compressor, the condenser, the expansion valve, and the evaporator are connected by piping so that refrigerant may circulate.

[0006] The indoor unit and the outdoor unit are disposed in separate locations. Therefore, after disposing the indoor unit and the outdoor unit, the indoor unit and the outdoor unit may be connected to each other using a connecting piping.

[0007] A service valve may be used to connect the connecting pipe connected to the indoor unit and the outdoor unit.

[0008] When the service valve is open, the refrigerant may flow through the piping. When the service valve is closed, the refrigerant may not flow through the piping. In other words, the service valve may control the flow of refrigerant circulating between the indoor unit and the outdoor unit.

SUMMARY

[0009] Certain example embodiments may provide a service valve for an air conditioner capable of preventing or reducing spindle wear.

[0010] According to an example embodiment, a service valve for air conditioner may include: a body; a fluid passage formed to penetrate the body; a spindle receiving part formed to communicate, directly or indirectly, with the fluid passage; a spindle screw-connected, directly or indirectly, to the spindle receiving part and configured to open and block the fluid passage; a guard hole formed at a certain depth on an upper surface of the spindle; and a wrench guard disposed in the guard hole.

[0011] The wrench guard may be formed in a sleeve shape including a through hole, and a cross-section of the through hole may be a regular hexagon.

[0012] The guard hole may be formed as a polygonal hole. An outer circumferential surface of the wrench guard may be formed as a polygon corresponding to the polygonal hole of the guard hole.

[0013] The polygonal hole may be formed as one of a square hole, a hexagonal hole, and a polygonal star hole.

[0014] The guard hole may be formed as an elliptical hole. An outer circumferential surface of the wrench guard may be formed as an elliptical shape corresponding to the elliptical hole of the guard hole.

[0015] The wrench guard may include a stopper provided at an upper end of the wrench guard. The wrench guard may be fixed to the guard hole by the stopper.

[0016] The wrench guard may include a fixing protrusion provided on an outer circumferential

surface of the wrench guard. The wrench guard may be fixed to the guard hole by the fixing protrusion.

[0017] The spindle may further include a wrench hole formed below the guard hole.

[0018] A length of the wrench guard may be equal to or greater than a depth of the wrench hole.

[0019] The length of the wrench guard may be at least 5 mm.

[0020] A material of the body may be brass, and a material of the wrench guard may be stainless steel.

[0021] The stainless steel may be STS 304 and/or STS 316, as examples.

[0022] The body may be formed in a cross shape. The fluid passage may be formed on an inside of a leg, a center portion, and a right arm of the body. The spindle receiving part may be formed to connect an upper surface of a head of the body **10** and the center portion.

[0023] The spindle receiving part may include a female screw portion. The spindle may include a screw portion fastened to the female screw portion.

[0024] The service valve may further include a charging core disposed in a left arm of the body.

Description

BRIEF DESCRIPTION OF DRAWINGS

[0025] These and/or other aspects, features, and advantages of certain embodiments of the disclosure will be more apparent from the following description taken in conjunction with the accompanying drawings, in which:

[0026] FIG. **1** is a perspective view illustrating a service valve **1** for an air conditioner according to an example embodiment.

[0027] FIG. **2** is a cross-sectional perspective view illustrating a service valve **1** for an air conditioner according to an example embodiment.

[0028] FIG. **3** is a perspective view illustrating a spindle **40** of a service valve **1** for an air conditioner according to an example embodiment.

[0029] FIG. **4** is a cross-sectional view illustrating the spindle **40** of FIG. **3**.

[0030] FIG. **5** is an exploded perspective view illustrating the spindle **40** of FIG. **3**.

[0031] FIG. **6** is a cross-sectional perspective view illustrating the spindle **40** of FIG. **3** without a wrench guard **50**.

[0032] FIG. **7** is a cross-sectional view illustrating a spindle **40** of a service valve **1** for an air conditioner according to an example embodiment.

[0033] FIG. **8** is a cross-sectional view illustrating a spindle **40** of a service valve **1** for an air conditioner according to an example embodiment.

[0034] FIG. **9** is a perspective view illustrating a spindle **40** of a service valve **1** for an air conditioner according to an example embodiment.

[0035] FIG. **10** is a cross-sectional view illustrating the spindle **40** of FIG. **9**.

[0036] FIG. **11** is an exploded perspective view illustrating the spindle **40** of FIG. **9**.

[0037] FIG. **12** is a cross-sectional perspective view illustrating the spindle **40** of FIG. **9** without a wrench guard **50**.

[0038] FIG. **13** is a perspective view illustrating a wrench guard **50** used in a spindle **40** of a service valve **1** for an air conditioner according to an example embodiment.

[0039] FIG. **14** is a perspective view illustrating the spindle **40** of a service valve **1** for an air conditioner according to an example embodiment, in which the wrench guard **50** of FIG. **13** is disposed.

[0040] FIG. **15** is a perspective view illustrating a wrench guard **50** used in a spindle **40** of a service valve **1** for an air conditioner according to an example embodiment.

[0041] FIG. **16** is a perspective view illustrating the spindle **40** of a service valve **1** for an air

conditioner according to an example embodiment, in which the wrench guard **50** of FIG. **15** is disposed.

[0042] FIG. **17** is a cross-sectional view illustrating a spindle **40** of a service valve **1** for an air conditioner according to an example embodiment.

[0043] FIG. **18** is a cross-sectional view illustrating a wrench guard **50** used in the spindle **40** of FIG. **17**.

[0044] FIG. **19** is a cross-sectional view illustrating a wrench guard **50** used in a spindle **40** of a service valve **1** for an air conditioner according to an example embodiment.

[0045] FIG. **20** is a cross-sectional view illustrating a spindle **40** of a service valve **1** for an air conditioner according to an example embodiment.

[0046] FIG. **21** is a cross-sectional perspective view illustrating a wrench guard **50** used in the spindle **40** of FIG. **20**.

[0047] FIG. **22** is a cross-sectional perspective view illustrating a wrench guard **50** used in a spindle **40** of a service valve **1** for an air conditioner according to an example embodiment.

[0048] FIG. **23** is a cross-sectional view illustrating a state in which a service valve **1** for an air conditioner according to an example embodiment is closed by a spindle **40**.

[0049] FIG. **24** is a cross-sectional view illustrating a state in which a service valve **1** for an air conditioner according to an example embodiment is opened by a spindle **40**.

[0050] FIG. **25** is a cross-sectional view illustrating a state in which a wrench is inserted into a wrench hole **42** of a spindle **40** of a service valve **1** for an air conditioner according to the prior art.

DETAILED DESCRIPTION

[0051] Since the embodiments of the disclosure can apply various transformations and have various embodiments, specific embodiments will be illustrated in the drawings and described in detail in the detailed description. However, this is not intended to limit the scope to the specific embodiments, and should be understood to include various modifications, equivalents, and/or alternatives of the example embodiment. In connection with the description of the drawings, like reference numerals may be used for like elements.

[0052] In describing the disclosure, when it is determined that a detailed description of a [0053] related known function or configuration may unnecessarily obscure the gist of the disclosure, a detailed description thereof will be omitted.

[0054] In addition, the following embodiments may be modified in many different forms, and the scope of the technical idea of the disclosure is not limited to the following embodiments. Rather, these embodiments are provided so that this disclosure will be thorough and complete, and will fully convey the spirit of the disclosure to those skilled in the art.

[0055] Terms used in this disclosure are only used to describe specific embodiments, and are not intended to limit the scope of rights. Singular expressions include plural expressions unless the context clearly dictates otherwise.

[0056] In this disclosure, expressions such as “has,” “can have,” “includes,” or “can include” indicate the existence of a corresponding feature (e.g., numerical value, function, operation, or component such as a part) and do not preclude the existence of additional features.

[0057] In this disclosure, expressions such as “A or B,” “at least one of A or/and B,” or “one or more of A or/and B” may include all possible combinations of the items listed together. For example, “A or B,” “at least one of A or/and B,” or “one or more of A or/and B” may refer to all cases (1) including at least one A, (2) including at least one B, or (3) including both at least one A and at least one B.

[0058] Expressions such as “first,” “second,” “primary,” or “secondary,” as used in this disclosure may modify various components regardless of order and/or importance, are used only to distinguish one component from other components, and do not limit the corresponding components.

[0059] Further, terms such as ‘leading end’, ‘rear end’, ‘upper side’, ‘lower side’, ‘top end’,

'bottom end', etc. used in the disclosure are defined with reference to the drawings. However, the shape and position of each component are not limited by these terms.

[0060] Hereinafter, embodiments of a service valve **1** for an air conditioner according to the disclosure will be described in detail with reference to the attached drawings.

[0061] FIG. **1** is a perspective view illustrating a service valve **1** for an air conditioner according to an example embodiment. FIG. **2** is a cross-sectional perspective view illustrating a service valve **1** for an air conditioner according to an example embodiment.

[0062] A service valve **1** for an air conditioner according to an example embodiment may include a body **10**, a fluid passage **20**, a spindle receiving part **30**, and a spindle **40**.

[0063] The body **10** may form the appearance of the service valve **1** for an air conditioner. The body **10** may be formed of brass.

[0064] The fluid passage **20** may be formed to penetrate the body **10**. In other words, two openings **20a** and **20b** connected to both ends of the fluid passage **20** may be provided on the surface of the body **10**.

[0065] The spindle receiving part **30** may be formed to communicate, directly or indirectly, with the fluid passage **20**. In other words, the spindle receiving part **30** may be formed to be connected, directly or indirectly, to the fluid passage **20** in the body **10**. The spindle receiving part **30** may be formed as a hole connecting the surface of the body **10** and the fluid passage **20**.

[0066] The spindle **40** may be disposed in the spindle receiving part **30**. The spindle **40** may be screw-fastened to the spindle receiving part **30**, and may be configured to open and close the fluid passage **20**.

[0067] Referring to FIGS. **1** and **2**, the body **10** may be formed in an approximately cross shape. In other words, the body **10** may include four branches extending perpendicularly from the center portion of the body **10**. The four branches may be formed in an approximately circular pipe shape.

[0068] A branch extending upward from the center portion **10a** of the body **10** may be referred to as a head **10b**. A branch extending downward from the center portion **10a** of the body **10** may be referred to as a leg **10c**. Branches extending right and left from the center portion **10a** of the body **10** may be referred to as a right arm **10d** and a left arm **10e**.

[0069] The right arm **10d** may be connected, directly or indirectly, to a connecting pipe **3** connected to an indoor unit. A flare nut **12** may be disposed at one end of the right arm **10d**. Accordingly, the right arm **10d** and the connecting pipe **3** may be connected by the flare nut **12**.

[0070] The leg **10c** may be connected, directly or indirectly, to a pipe **4** of an outdoor unit.

[0071] The fluid passage **20** may be formed inside the leg **10c**, the center portion **10a**, and the right arm **10d**. In other words, the fluid passage **20** may be formed to penetrate the leg **10c**, the center portion **10a**, and the right arm **10d** of the body **10**. Accordingly, refrigerant may flow through the fluid passage **20** formed inside the body **10**.

[0072] The fluid passage **20** may include a first fluid passage **21** formed in the longitudinal direction of the leg **10c**, a second fluid passage **22** formed in the longitudinal direction of the right arm **10d**, and a central space **23** formed in the center portion **10a** and connecting the first fluid passage **21** and the second fluid passage **22**. In other words, the fluid passage **20** may be formed by the first fluid passage **21**, the central space **23**, and the second fluid passage **22**.

[0073] A charging passage **14** communicating with the center portion **10a** may be provided inside the left arm **10e**. A charging core **15** may be disposed at one end of the left arm **10e**. The refrigerant may be injected into the fluid passage **20** using the charging core **15**. A core cap **13** may be provided at one end of the left arm **10e**.

[0074] The spindle receiving part **30** may be formed in the head **10b** of the body **10** to be in communication with the center portion **10a**. The spindle receiving part **30** may be formed to connect the upper surface of the head **10b** and the center portion **10a**. In other words, the spindle receiving part **30** may be in communication with the central space **23** of the center portion **10a**.

[0075] A female screw portion **31** may be provided on the inner surface of the spindle receiving

part **30**. The female screw portion **31** may be provided so as to be adjacent to the fluid passage **20**. The female screw portion **31** of the spindle receiving part **30** may be formed so that a screw portion **44** of the spindle **40** may be fastened.

[0076] The spindle **40** may be disposed in the spindle receiving part **30**. The spindle **40** may be disposed so as to move up and down with respect to the spindle receiving part **30**. The spindle **40** may include the screw portion **44**. The screw portion **44** may be formed as a male screw. The screw portion **44** of the spindle **40** may be formed as a male screw corresponding to the female screw portion **31** of the spindle receiving part **30**. Therefore, the spindle **40** may be screw-coupled to the inside of the spindle receiving part **30**.

[0077] Because the spindle **40** and the spindle receiving part **30** are screw-connected, directly or indirectly, the rotational motion of the spindle **40** may be converted into the linear motion of the spindle **40**.

[0078] For example, when the spindle **40** is rotated, the spindle **40** may move linearly up and down inside the spindle receiving part **30**. In detail, when the spindle **40** is rotated in one direction, the spindle **40** may move downward in the spindle receiving part **30**. When the spindle **40** is rotated in the opposite direction, the spindle **40** may move upward in the spindle receiving part **30**.

[0079] A spindle seat **24** corresponding to the lower end of the spindle **40** may be provided at the lower end of the center portion **10a** of the body **10**, that is, at the upper end of the leg **10c**. The spindle seat **24** may be formed as a chamfer at the upper end of the first fluid passage **21**. In other words, the spindle seat **24** may be provided below the spindle receiving part **30**.

[0080] A chamfer **45** corresponding to the spindle seat **24** may be provided at the lower end of the spindle **40**. Accordingly, when the chamfer **45** at the lower end of the spindle **40** comes into contact with the spindle seat **24**, the refrigerant may not flow between the lower end of the spindle **40** and the spindle seat **24**. In other words, when the lower end of the spindle **40** is in contact with the spindle seat **24**, the first fluid passage **21** may be blocked. Then, the fluid passage **20** may be blocked so that the refrigerant may not flow through the fluid passage **20**.

[0081] When the spindle **40** moves upward and the lower end of the spindle **40** is not in contact with the spindle seat **24**, the refrigerant may flow between the lower end of the spindle **40** and the spindle seat **24**.

[0082] At least one O-ring **35** may be disposed on the upper portion of the spindle **40**. In detail, at least one O-ring groove **47** may be provided on the upper portion of the spindle **40**, that is, above the screw portion **44**. An O-ring **35** may be disposed in the at least one O-ring groove **47**. The O-ring **35** may prevent the refrigerant from leaking between the spindle **40** and the spindle receiving part **30**.

[0083] In this embodiment, two O-ring grooves **47** and two O-rings **35** disposed in the two O-ring grooves **47** may be provided at the upper portion of the spindle **40**.

[0084] A stop ring **32** may be disposed on the upper end of the spindle receiving part **30**. The stop ring **32** may be provided to prevent the spindle **40** from falling out of the spindle receiving part **30**. Accordingly, when high pressure is applied to the fluid passage **20**, the spindle **40** may not fall out of the spindle receiving part **30**.

[0085] Hereinafter, the spindle **40** according to an example embodiment will be described in detail with reference to FIGS. **3** to **6**.

[0086] FIG. **3** is a perspective view illustrating a spindle **40** of a service valve **1** for an air conditioner according to an example embodiment. FIG. **4** is a cross-sectional view illustrating the spindle **40** of FIG. **3**. FIG. **5** is an exploded perspective view illustrating the spindle **40** of FIG. **3**. FIG. **6** is a cross-sectional perspective view illustrating the spindle **40** of FIG. **3** without a wrench guard **50**.

[0087] Referring to FIGS. **3** to **6**, the spindle **40** of the service valve **1** for an air conditioner according to an example embodiment may include a guard hole **41**, a wrench guard **50**, a screw portion **44**, and an O-ring groove **47**.

[0088] The guard hole **41** may be formed at a certain depth on the upper surface of the spindle **40**. The guard hole **41** may be formed so that the wrench guard **50** is inserted into the guard hole **41**. The spindle **40** may be formed of a brass material.

[0089] The guard hole **41** may be formed in a shape corresponding to the wrench guard **50**. In detail, the cross-section of the guard hole **41** may be formed identically to the cross-section of the wrench guard **50**. For example, the guard hole **41** may be formed as a polygonal hole.

[0090] In the case of this embodiment, because the cross-section of the outer circumferential surface **52** of the wrench guard **50** is formed as a regular hexagon, the cross-section of the guard hole **41** may be formed as a regular hexagon.

[0091] The guard hole **41** may be formed to have a depth that firmly and stably supports the wrench guard **50**. The depth of the guard hole **41** may be defined according to the length of the wrench guard **50**.

[0092] A step **411** may be provided at the bottom of the guard hole **41** to limit the insertion depth of the wrench guard **50**. The step may be formed to protrude from the inner surface of the guard hole **41** to a certain height toward the center of the guard hole **41**.

[0093] In the case where the wrench hole **42** is formed at the lower side of the guard hole **41**, the upper end of the wrench hole **42** may form the step **411**. In the case where there is only a drill groove **43** below the guard hole **41**, the upper end of the drill groove **43** may form the step **411**.

[0094] The wrench guard **50** may be disposed in the guard hole **41**. The wrench guard **50** may be formed in a sleeve shape. In other words, the wrench guard **50** may have a through hole **51** formed in the center of the wrench guard **50**. Therefore, the wrench guard **50** may include an inner circumferential surface and an outer circumferential surface. For example, the wrench guard **50** may be formed in a sleeve shape in which the cross-section of the inner circumferential surface of the through hole **51** is a regular hexagon.

[0095] The inner circumferential surface of the through hole **51** of the wrench guard **50** may be formed in a shape corresponding to a hexagonal wrench (Hex key). The hexagonal wrench refers to a tool used for removing and fastening a bolt or screw (Hex socket head cap screw) having a regular hexagonal groove formed on the head of the bolt. For example, in the case of an M4 hexagonal wrench, the through hole **51** of the wrench guard **50** may be formed in a regular hexagonal cross-section corresponding to the cross-section of the M4 hexagonal wrench.

[0096] The outer circumferential surface **52** of the wrench guard **50** may be formed in a shape corresponding to the guard hole **41**. In detail, the cross-section of the outer circumferential surface **52** of the wrench guard **50** may be formed in a shape corresponding to the cross-section of the guard hole **41**. In the case of this embodiment, because the cross-section of the guard hole **41** is formed in a regular hexagon, the cross-section of the outer circumferential surface **52** of the wrench guard **50** may be formed in a regular hexagon.

[0097] In other words, the wrench guard **50** according to this embodiment may be formed with the inner circumferential surface having a regular hexagonal cross-section and the outer circumferential surface **52** having a regular hexagonal cross-section.

[0098] The wrench guard **50** may be formed of stainless steel. The stainless steel may be either STS 304 or STS 316 (KS standard).

[0099] When the wrench guard **50** is formed of stainless steel, the spindle **40** made of brass may be prevented from being worn by the hexagonal wrench. In addition, when the wrench guard **50** is formed of stainless steel, the wrench guard **50** may be prevented from being corroded.

[0100] The wrench guard **50** may be formed to have a length that allows the hexagonal wrench to apply sufficient force to the spindle **40** to rotate the spindle **40** when rotating the spindle **40** with the hexagonal wrench.

[0101] For example, as illustrated in FIG. 25, the spindle **40** may be formed of brass, and a regular hexagonal groove corresponding to a hexagonal wrench **100**, that is, a wrench hole **42**, may be formed on the upper surface of the spindle **40**. In this case, the depth of the wrench hole **42**

required to turn the spindle **40** without the wrench guard **50** with the hexagonal wrench **100** may be L. In other words, the depth of the brass wrench hole **42** is L. As in this embodiment, when the wrench guard **50** is disposed in the spindle **40** made of brass, the length of the wrench guard **50** may be L/2 or more.

[0102] In addition, the wrench guard **50** may be formed of thin stainless steel. The wrench guard **50** may be formed to have a thickness that prevents the wrench guard **50** from being damaged by the hexagonal wrench **100** when the spindle **40** is rotated with the hexagonal wrench **100**. For example, the wrench guard **50** may be formed of stainless steel having a thickness of 0.3 mm to 0.5 mm.

[0103] Referring to FIGS. **4** and **6**, the spindle **40** according to an example embodiment may include a guard hole **41** and a wrench hole **42**.

[0104] The wrench hole **42** may be formed below the guard hole **41**. The wrench hole **42** may be formed in a shape corresponding to the hexagonal wrench. In other words, the wrench hole **42** may be formed in a shape identical to the cross-section of the through hole **51** of the wrench guard **50**. The wrench hole **42** may be formed in a regular hexagonal shape identical to the through hole **51** of the wrench guard **50**.

[0105] The depth of the wrench hole **42** may be formed identical to the depth of the guard hole **41**. Alternatively, the depth of the wrench hole **42** may be formed shorter than the depth of the guard hole **41**.

[0106] In the case where the wrench hole **42** is formed in the spindle **40**, when the spindle **40** is turned with the hexagonal wrench, the hexagonal wrench may be inserted into the through hole **51** of the wrench guard **50** and the wrench hole **42**. Therefore, the hexagonal wrench may contact the inner circumferential surface of the wrench guard **50** and the inner circumferential surface of the wrench hole **42**. Accordingly, the hexagonal wrench may stably rotate the spindle **40** and prevent the inner surface of the spindle **40** from being worn by the hexagonal wrench.

[0107] A drill groove **43** for forming the wrench hole **42** may be provided below the wrench hole **42**.

[0108] As another example, as illustrated in FIG. **7**, the wrench hole **42** may not be formed below the guard hole **41**.

[0109] FIG. **7** is a cross-sectional view illustrating a spindle **40** of a service valve **1** for an air conditioner according to an example embodiment.

[0110] Referring to FIG. **7**, the guard hole **41** may be formed in the upper surface of the spindle **40**, and the wrench guard **50** may be disposed in the guard hole **41**. The depth of the guard hole **41** may be formed to be the same as the length of the wrench guard **50**. The length of the wrench guard **50** may be approximately half the depth of the brass wrench hole **42**. In other words, when the depth of the brass wrench hole **42** is L, the length of the wrench guard **50** may be L/2.

[0111] A drill groove **43** for forming the guard hole **41** may be provided below the guard hole **41**.

[0112] FIG. **8** is a cross-sectional view illustrating a spindle **40** of a service valve **1** for an air conditioner according to an example embodiment.

[0113] Referring to FIG. **8**, the guard hole **41** may be formed in the upper surface of the spindle **40**, and the wrench guard **50** may be disposed in the guard hole **41**. The depth of the guard hole **41** may be formed to be the same as the length of the wrench guard **50**. The length of the wrench guard **50** may be the same as the depth of the brass wrench hole **42**. In other words, when the depth of the brass wrench hole **42** is L, the length of the wrench guard **50** may be L.

[0114] The drill groove **43** for forming the guard hole **41** may be provided below the guard hole **41**.

[0115] The screw portion **44** may be formed on the outer circumferential surface of the spindle **40**. The screw portion **44** may be formed on the lower portion of the outer circumferential surface of the spindle **40**. The screw portion **44** may be formed as a male screw. The screw portion **44** may be formed as a male screw corresponding to the female screw portion **31** of the spindle receiving part **30**. Therefore, the spindle **40** may be screw-coupled to the inside of the spindle receiving part **30**.

[0116] Therefore, when the hexagonal wrench is inserted into the through hole **51** of the wrench

guard **50** and the spindle **40** is rotated with the hexagonal wrench, the spindle **40** may move up and down inside the spindle receiving part **30**.

[0117] The chamfer **45** corresponding to the chamfer of the spindle seat **24** may be provided at the lower end of the spindle **40**, that is, below the screw portion **44**.

[0118] The O-ring groove **47** may be formed on the outer circumferential surface of the spindle **40**. The O-ring groove **47** may be formed on the upper portion of the outer circumferential surface of the spindle **40**. In other words, the O-ring groove **47** may be provided above the screw portion **44** on the outer circumferential surface of the spindle **40**. The O-ring **35** (see FIG. 2) may be disposed in the O-ring groove **47**. The O-ring **35** may prevent refrigerant from leaking between the spindle **40** and the spindle receiving part **30**.

[0119] In this embodiment, two O-ring grooves **47** are formed on the upper portion of the outer circumferential surface of the spindle **40**. However, the number of O-ring grooves **47** formed on the outer circumferential surface of the spindle **40** may not be limited thereto. One or three or more O-ring grooves **47** may be provided on the spindle **40**.

[0120] In the above, the case where the outer circumferential surface **52** of the wrench guard **50** is formed as a regular hexagon has been described, but the disclosure is not limited thereto. The outer circumferential surface **52** of the wrench guard **50** used in the spindle **40** according to the disclosure may be formed in various shapes as long as the wrench guard **50** can be prevented from rotating in the guard hole **41** of the spindle **40**.

[0121] For example, the outer circumferential surface **52** of the wrench guard **50** may be formed as a polygon corresponding to the guard hole **41** formed as a polygonal hole.

[0122] Hereinafter, a wrench guard **50** having a rectangular outer circumferential surface will be described with reference to FIGS. 9 to 12.

[0123] FIG. 9 is a perspective view illustrating a spindle **40** of a service valve **1** for an air conditioner according to an example embodiment. FIG. 10 is a cross-sectional view illustrating the spindle **40** of FIG. 9. FIG. 11 is an exploded perspective view illustrating the spindle **40** of FIG. 9. FIG. 12 is a cross-sectional perspective view illustrating the spindle **40** of FIG. 9 without a wrench guard **50**.

[0124] Referring to FIGS. 9 to 12, a spindle **40** of the service valve **1** for an air conditioner according to an example embodiment may include a guard hole **41**, a wrench guard **50**, a screw portion **44**, and an O-ring groove **47**.

[0125] The guard hole **41** may be formed at a certain depth in the upper surface of the spindle **40**. The guard hole **41** may be formed so that the wrench guard **50** is inserted into the guard hole **41**. The spindle **40** may be formed of a brass material.

[0126] The guard hole **41** may be formed in a shape corresponding to the wrench guard **50**. In detail, the cross-section of the guard hole **41** may be formed identically to the cross-section of the wrench guard **50**. For example, the guard hole **41** may be formed as a polygonal hole.

[0127] In the case of this embodiment, because the cross-section of the outer circumferential surface **52** of the wrench guard **50** is formed as a square, the cross-section of the guard hole **41** may be formed as a square. In other words, the guard hole **41** may be formed as a square hole.

[0128] The guard hole **41** may be formed to have a depth that firmly and stably supports the wrench guard **50**. The depth of the guard hole **41** may be defined according to the length of the wrench guard **50**.

[0129] The wrench guard **50** may be disposed in the guard hole **41**. The wrench guard **50** may be formed in a sleeve shape. In other words, the wrench guard **50** may have a through hole **51** formed in the center of the wrench guard **50**. Therefore, the wrench guard **50** may include an inner circumferential surface and an outer circumferential surface. For example, the wrench guard **50** may be formed in a sleeve shape in which the cross-section of the through hole **51**, that is, the cross-section of the inner circumferential surface thereof is a regular hexagon and the cross-section of the outer circumferential surface thereof is a square.

[0130] The through hole **51** of the wrench guard **50** may be formed in the same manner as the through hole **51** of the wrench guard **50** according to the above-described embodiment. In other words, the through hole **51** of the wrench guard **50** may be formed in a shape corresponding to the hexagonal wrench (Hex key).

[0131] The outer circumferential surface **52** of the wrench guard **50** may be formed in a shape corresponding to the guard hole **41**. In the case of this embodiment, because the cross-section of the guard hole **41** is formed in a square shape, the cross-section of the outer circumferential surface **52** of the wrench guard **50** may be formed in a square shape.

[0132] In other words, the wrench guard **50** according to this embodiment may include the inner circumferential surface having a regular hexagonal cross-section and the outer circumferential surface having a square cross-section.

[0133] The wrench guard **50** may be formed of stainless steel. The stainless steel may be either STS 304 or STS 316 (KS standard).

[0134] The wrench guard **50** may be formed to have a length that allows the hexagonal wrench to apply sufficient force to the spindle **40** to rotate the spindle **40** when the hexagonal wrench is used to rotate the spindle **40**.

[0135] In this embodiment, the length of the wrench guard **50** may be the same as the depth of the brass wrench hole **42**. In other words, when the depth of the brass wrench hole **42** is L, the length of the wrench guard **50** may be L.

[0136] A drill groove **43** may be provided below the guard hole **41** to form the guard hole **41**.

[0137] The screw portion **44** and the O-ring groove **47** are the same as the screw portion **44** and the O-ring groove **47** according to the above-described embodiment, so a detailed descriptions thereof are omitted.

[0138] Hereinafter, a wrench guard **50** having an outer circumferential surface in the shape of a hexagonal star will be described with reference to FIGS. **13** and **14**.

[0139] FIG. **13** is a perspective view illustrating a wrench guard **50** used in a spindle **40** of a service valve **1** for an air conditioner according to an example embodiment. FIG. **14** is a perspective view illustrating the spindle **40** of a service valve **1** for an air conditioner according to an example embodiment, in which the wrench guard **50** of FIG. **13** is disposed.

[0140] Referring to FIG. **13**, the wrench guard **50** may be formed in a sleeve shape in which the cross-section of the through hole **51** is a regular hexagon and the cross-section of the outer circumferential surface **52** is a hexagonal star shape.

[0141] The through hole **51** of the wrench guard **50** may be formed in the same manner as the through hole **51** of the wrench guard **50** according to the above-described embodiment. In other words, the through hole **51** of the wrench guard **50** may be formed in a shape corresponding to the hexagonal wrench.

[0142] The cross-section of the outer circumferential surface **52** of the wrench guard **50** may be formed in a hexagonal star shape. In other words, the outer circumferential surface **52** of the wrench guard **50** may be formed in a hexagonal star pillar shape.

[0143] Therefore, the wrench guard **50** according to this embodiment may include the inner circumferential surface having a regular hexagonal cross-section and the outer circumferential surface having a hexagonal star-shaped cross-section.

[0144] The wrench guard **50** may be disposed in the guard hole **41** of the spindle **40**.

[0145] Referring to FIG. **14**, the guard hole **41** may be formed in the upper surface of the spindle **40**. The guard hole **41** may be formed in a shape corresponding to the outer circumferential surface **52** of the wrench guard **50**. In the case of this embodiment, because the cross-section of the outer circumferential surface **52** of the wrench guard **50** is formed in the hexagonal star shape, the cross-section of the guard hole **41** may be formed in the hexagonal star shape. In other words, the guard hole **41** may be formed as a hexagonal star hole.

[0146] Therefore, when the wrench guard **50** is inserted into the guard hole **41** of the spindle **40**,

the wrench guard **50** may not rotate relative to the guard hole **41**.

[0147] The guard hole **41** may be formed to have a depth that firmly and stably supports the wrench guard **50**. The depth of the guard hole **41** may be defined according to the length of the wrench guard **50**.

[0148] In FIGS. **13** and **14**, the cross-section of the outer circumferential surface **52** of the wrench guard **50** and the cross-section of the guard hole **41** are illustrated and described as having a hexagonal star shape. However, the cross-section of the outer circumferential surface **52** of the wrench guard **50** and the cross-section of the guard hole **41** are not limited to the hexagonal star shape. The cross-section of the outer circumferential surface **52** of the wrench guard **50** and the cross-section of the guard hole **41** may be formed in a polygonal star shape such as a pentagonal star, a heptagonal star, an octagonal star, etc.

[0149] In the above, the case where the cross-section of the outer circumferential surface **52** of the wrench guard **50** is a square or polygonal star shape has been described, but the wrench guard **50** according to the disclosure is not limited thereto. The cross-section of the outer circumferential surface **52** of the wrench guard **50** may be formed as a polygon greater than a triangle, a pentagon, or a hexagon or more polygonal shape. In addition, the cross-section of the outer circumferential surface **52** of the wrench guard **50** may be formed in an elliptical shape.

[0150] Hereinafter, a wrench guard **50** having an elliptical outer circumferential surface will be described with reference to FIGS. **15** and **16**.

[0151] FIG. **15** is a perspective view illustrating a wrench guard **50** used in a spindle **40** of a service valve **1** for an air conditioner according to an example embodiment. FIG. **16** is a perspective view illustrating the spindle **40** of the service valve **1** for an air conditioner according to an example embodiment, in which the wrench guard **50** of FIG. **15** is disposed.

[0152] Referring to FIG. **15**, the wrench guard **50** may be formed in a sleeve shape in which the cross-section of the through hole **51** is a regular hexagon and the cross-section of the outer circumferential surface **52** is an elliptical shape.

[0153] The through hole **51** of the wrench guard **50** may be formed in the same manner as the through hole **51** of the wrench guard **50** according to the above-described embodiment. In other words, the through hole **51** of the wrench guard **50** may be formed in a shape corresponding to the hexagonal wrench.

[0154] The cross-section of the outer circumferential surface of the wrench guard **50** may be formed in an elliptical shape. In other words, the outer circumferential surface of the wrench guard **50** may be formed in an elliptical pillar shape.

[0155] Therefore, the wrench guard **50** according to this embodiment may include the inner circumferential surface having a regular hexagonal cross-section and the outer circumferential surface **52** having an elliptical cross-section.

[0156] Referring to FIG. **16**, the guard hole **41** may be formed in the upper surface of the spindle **40**. The guard hole **41** may be formed in a shape corresponding to the outer circumferential surface **52** of the wrench guard **50**. In the case of this embodiment, because the cross-section of the outer circumferential surface **52** of the wrench guard **50** is formed in the elliptical shape, the cross-section of the guard hole **41** may be formed in the elliptical shape. In other words, the guard hole **41** may be formed as an elliptical hole.

[0157] Therefore, when the wrench guard **50** is inserted into the guard hole **41** of the spindle **40**, the wrench guard **50** may not rotate relative to the guard hole **41**.

[0158] The guard hole **41** may be formed to have a depth that firmly and stably supports the wrench guard **50**. The depth of the guard hole **41** may be defined according to the length of the wrench guard **50**.

[0159] The wrench guard **50** may include a stopper so as not to fall out of the guard hole **41**. Hereinafter, a wrench guard **50** having a stopper will be described with reference to FIGS. **17** and **18**.

[0160] FIG. 17 is a cross-sectional view illustrating a spindle 40 of a service valve 1 for an air conditioner according to an example embodiment. FIG. 18 is a cross-sectional view illustrating a wrench guard 50 used in the spindle 40 of FIG. 17.

[0161] Referring to FIGS. 17 and 18, the wrench guard 50 may include a stopper 55.

[0162] The stopper 55 may be provided at the upper end of the wrench guard 50. The stopper 55 may be formed to extend radially from the upper end of the wrench guard 50. In other words, the stopper 55 may be formed in a flange shape at the upper end of the wrench guard 50.

[0163] The stopper 55 may be formed in the same shape as the outer circumferential surface 52 of the wrench guard 50. For example, when the outer circumferential surface 52 of the wrench guard 50 is formed in a regular hexagon, the stopper 55 may be formed in a regular hexagon.

Alternatively, the stopper 55 may be formed in a circular shape.

[0164] The thickness t_1 of the stopper 55 may be formed to be less than or equal to the thickness t of the wrench guard 50. The upper surface 551 of the stopper 55 and the lower surface 552 of the stopper 55 may be formed parallel to each other. In other words, the cross-section of the stopper 55 may be formed in a rectangular shape.

[0165] A stopper seat 46 may be provided at the upper end of the guard hole 41 of the spindle 40. When the wrench guard 50 is inserted into the guard hole 41, the stopper 55 of the wrench guard 50 may be received in the stopper seat 46 of the guard hole 41.

[0166] Because the stopper 55 of the wrench guard 50 is forcibly fitted into the stopper seat 46 of the guard hole 41, the wrench guard 50 may not fall out of the guard hole 41 of the spindle 40.

[0167] In the above-described embodiment, the lower surface 552 of the stopper 55 is formed parallel to the upper surface 551 of the stopper 55. However, the lower surface 552 of the stopper 55 may be formed inclined with respect to the upper surface 551 of the stopper 55, as illustrated in FIG. 19.

[0168] FIG. 19 is a cross-sectional view illustrating a wrench guard 50 used in a spindle 40 of a service valve 1 for an air conditioner according to an example embodiment.

[0169] Referring to FIG. 19, the upper surface 551 of the stopper 55 may be formed parallel to the lower surface of the wrench guard 50, and the lower surface 552 of the stopper 55 may be formed inclined upward with respect to the upper surface 551 of the stopper 55. Therefore, the cross-section of the stopper 55 may be formed in a right-angled triangle shape.

[0170] The wrench guard 50 may include a fixing protrusion to prevent the wrench guard 50 from falling out of the guard hole 41. Hereinafter, a wrench guard 50 having a fixing protrusion will be described with reference to FIGS. 20 and 21.

[0171] FIG. 20 is a cross-sectional view illustrating a spindle 40 of a service valve 1 for an air conditioner according to an example embodiment. FIG. 21 is a cross-sectional perspective view illustrating a wrench guard 50 used in the spindle 40 of FIG. 20.

[0172] Referring to FIGS. 20 and 21, the wrench guard 50 may include a fixing protrusion 56.

[0173] The fixing protrusion 56 may be formed on the outer circumferential surface of the wrench guard 50. The fixing protrusion 56 may be formed to protrude outward from the outer circumferential surface 52 of the wrench guard 50. For example, when the outer circumferential surface 52 of the wrench guard 50 is formed as a regular hexagonal column, the fixing protrusion 56 may be formed to protrude from the outer surface of the regular hexagonal column.

[0174] The fixing protrusion 56 may be positioned below the upper end of the wrench guard 50. The fixing protrusion 56 may be positioned on the upper portion of the wrench guard 50. In other words, the fixing protrusion 56 may be positioned above the center of the wrench guard 50 in the longitudinal direction of the wrench guard 50.

[0175] Two or more fixing protrusions 56 may be formed. Two or more fixing protrusions 56 may be symmetrically provided on the outer circumferential surface 52 of the wrench guard 50.

[0176] The fixing protrusion 56 may be formed to have a downwardly inclined surface 562. For example, the fixing protrusion 56 may be formed in a triangular cross-section. In the case where the

fixing protrusion **56** has the downwardly inclined surface **562**, when the wrench guard **50** equipped with the fixing protrusion **56** is inserted into the guard hole **41** of the spindle **40**, the wrench guard **50** may be easily inserted into the guard hole **41**.

[0177] The wrench guard **50** may be formed of thin stainless steel. For example, the wrench guard **50** may be formed of stainless steel having a thickness of 0.3 mm to 0.5 mm. Therefore, when a plurality of fixing protrusions **56** are provided on the outer circumferential surface **52** of the wrench guard **50**, the wrench guard **50** may be easily inserted into the guard hole **41** of the spindle **40** due to elastic deformation of the wrench guard **50**.

[0178] Because the spindle **40** is formed of brass and the fixing protrusion **56** of the wrench guard **50** is formed of stainless steel, the fixing protrusion **56** of the wrench guard **50** may be inserted into and fixed on the inner circumferential surface of the guard hole **41** of the spindle **40**.

[0179] The fixing protrusion **56** may be formed by molding the wrench guard **50** as illustrated in FIG. **22**.

[0180] FIG. **22** is a cross-sectional perspective view illustrating a wrench guard **50** used in a spindle **40** of a service valve **1** for an air conditioner according to an example embodiment.

[0181] Referring to FIG. **22**, the wrench guard **50** may include a fixing protrusion **56**. The fixing protrusion **56** may be formed by molding the wrench guard **50**.

[0182] For example, because the wrench guard **50** is formed of thin stainless steel, a force may be applied to the outside from one point on the inner circumferential surface of the wrench guard **50** to form the fixing protrusion **56**. Then, a concave groove **563** may be formed on the inner circumferential surface of the wrench guard **50**, and an externally convex fixing protrusion **56** may be formed on the outer circumferential surface **52** of the wrench guard **50**.

[0183] The fixing protrusion **56** may be positioned below the upper end of the wrench guard **50**. The fixing protrusion **56** may be positioned on the upper portion of the wrench guard **50**. In other words, the fixing protrusion **56** may be positioned above the center of the wrench guard **50** in the longitudinal direction of the wrench guard **50**.

[0184] Two or more fixing protrusions **56** may be formed. Two or more fixing protrusions **56** may be symmetrically provided on the outer circumferential surface **52** of the wrench guard **50**.

[0185] The fixing protrusion **56** may be formed to have a downwardly inclined surface **562**. For example, the fixing protrusion **56** may be formed in a triangular cross-section.

[0186] Hereinafter, the operation of the spindle **40** of the service valve **1** for an air conditioner according to an example embodiment will be described with reference to FIGS. **23** and **24**.

[0187] FIG. **23** is a cross-sectional view illustrating a state in which a service valve **1** for an air conditioner according to an example embodiment is closed by a spindle **40**.

[0188] Referring to FIG. **23**, the spindle **40** may be disposed in the spindle receiving part **30** of the service valve **1**. At this time, the male screw of the screw portion **44** of the spindle **40** is fastened to the female screw portion **31** of the spindle receiving part **30**.

[0189] The lower end of the spindle **40** may contact the spindle seat **24** provided at the upper end of the first fluid passage **21**. In other words, the chamfer **45** of the lower end of the spindle **40** may contact the spindle seat **24** of the body **10**. Therefore, the fluid passage **20** provided in the body **10** may be blocked by the spindle **40**.

[0190] In this state, when a hexagonal wrench is inserted into the through hole **51** of the wrench guard **50** of the spindle **40** and the hexagonal wrench is turned, the spindle **40** may rotate. When the hexagonal wrench is turned in one direction, the spindle **40** may move upward in the spindle receiving part **30**.

[0191] When the spindle **40** moves upward, the lower end of the spindle **40** is separated from the spindle seat **24** of the first fluid passage **21**, so that the first fluid passage **21** and the second fluid passage **22** may be connected, directly or indirectly, to each other.

[0192] When the hexagonal wrench is continuously turned, the spindle **40** may rise and the fluid passage **20** may be opened to the maximum as illustrated in FIG. **24**.

[0193] FIG. 24 is a cross-sectional view illustrating a state in which a service valve 1 for an air conditioner according to an example embodiment is opened by a spindle 40.

[0194] Referring to FIG. 24, when the service valve 1 is fully opened, the upper end of the spindle 40 may be adjacent to the upper end of the spindle receiving part 30, and the lower end of the spindle 40 may be maximally spaced from the spindle seat 24.

[0195] Then, the refrigerant (arrow F) may circulate between the outdoor unit and the indoor unit through the fluid passage 20 formed with the first fluid passage 21, the central space 23, and the second fluid passage 22.

[0196] In FIG. 24, when the hexagonal wrench is inserted into the through hole 51 of the wrench guard 50 of the spindle 40 and the hexagonal wrench is rotated in the opposite direction, the spindle 40 may move downward in the spindle receiving part 30.

[0197] When the spindle 40 moves downward and the chamfer 45 of the lower end of the spindle 40 comes into contact with the spindle seat 24, the fluid passage 20 may be blocked by the spindle 40. In other words, as illustrated in FIG. 23, the service valve 1 may be closed by the spindle 40.

[0198] FIG. 25 is a cross-sectional view illustrating a state in which a hexagonal wrench 100 is inserted into a wrench hole 42 of a spindle 40 of a service valve 1 for an air conditioner according to the prior art.

[0199] Referring to FIG. 25, a wrench hole 42 may be provided on the upper surface of a spindle 40. The wrench hole 42 may be formed in a shape corresponding to the hexagonal wrench 100. In other words, because the hexagonal wrench 100 has a regular hexagonal cross-section, the wrench hole 42 may be formed in a regular hexagonal cross-section.

[0200] The wrench hole 42 may be formed to have a depth that allows the hexagonal wrench 100 to apply sufficient force to the spindle 40 to rotate the spindle 40 when the spindle 40 is rotated with the hexagonal wrench 100 inserted into the wrench hole 42. When the spindle 40 is formed of a brass material, the depth of the wrench hole 42 required to rotate the spindle 40 without damaging the wrench hole 42 with the hexagonal wrench 100 may be L. In other words, the depth of the brass wrench hole 42 may be L.

[0201] In the case where the wrench hole 42 is formed of brass, when the hexagonal wrench 100 is not fully inserted into the wrench hole 42, the inner surface of the wrench hole 42 may be worn by the hexagonal wrench 100 when the spindle 40 is rotated with the hexagonal wrench 100.

Generally, the hexagonal wrench 100 is formed of stainless steel, steel, or the like, which has greater strength than brass, so when the hexagonal wrench 100 is not fully inserted into the wrench hole 42 of the spindle 40, the wrench hole 42 may be damaged by the hexagonal wrench 100.

[0202] However, because the service valve 1 according to an example embodiment has the wrench guard 50 formed of stainless steel disposed on the upper surface of the spindle 40, when the hexagonal wrench 100 is not fully inserted into the wrench guard 50, the wrench guard 50 may not be damaged by the hexagonal wrench 100.

[0203] In addition, when the wrench guard 50 of the service valve 1 according to an example embodiment is formed to have a length that is half the length of the brass wrench hole 42, the spindle 40 may be easily rotated with the hexagonal wrench 100.

[0204] In the case of closing the fluid passage 20 with the spindle 40, when the spindle 40 is rotated excessively, the lower end of the spindle 40 and the spindle seat 24 may be bonded by pressure. When the spindle 40 is bonded by pressure in this way, in the case where the wrench hole 42 of the spindle 40 is formed of brass as in the prior art, when the spindle 40 is rotated in the opposite direction using the hexagonal wrench 100, the bonding by pressure may not be released and the wrench hole 42 of the spindle 40 may be worn by the hexagonal wrench 100. In other words, the service valve 1 according to the prior art has a problem in that the spindle 40 may not be released when the spindle 40 is bonded by pressure.

[0205] However, in the case where the wrench guard 50 is disposed on the upper portion of the spindle 40, such as in the service valve 1 according to an example embodiment, when the spindle

40 is bonded by pressure, the spindle **40** may be rotated by inserting the hexagonal wrench **100** into the through hole **51** of the wrench guard **50** and turning the hexagonal wrench **10** in the opposite direction to release the bonding by pressure of the spindle **40**.

[0206] In the case of this disclosure, because the wrench guard **50** to which the hexagonal wrench **100** applies a force is formed of stainless steel, a force greater than the bonding force generated by the bonding by pressure between the lower end of the spindle **40** and the spindle seat **24** formed of brass may be applied to the wrench guard **50**. Therefore, the service valve **1** according to an example embodiment may release the spindle **40** when the spindle **40** is bonded by pressure.

[0207] According to an example embodiment, the service valve **1** may form the spindle **40** itself from stainless steel. In this case, there is no need to dispose the wrench guard **50** formed of stainless steel on the upper surface of the spindle **40**. Therefore, only a wrench hole **42** may be formed on the upper surface of the spindle **40** without the guard hole **41**.

[0208] When the spindle **40** is rotated by inserting the hexagonal wrench **100** into the wrench hole **42**, the wrench hole **42** of the spindle **40** formed from stainless steel may not wear out due to the hexagonal wrench **100**.

[0209] In addition, because a large force may be applied to the wrench hole **42** with the hexagonal wrench **100**, the spindle **40** may be released when the bonding by pressure of the spindle **40** occurs.

[0210] In addition, when the lower end of the spindle **40** is positioned on the spindle seat **24** of the body **10**, the contact between the lower end of the spindle **40** formed of stainless steel and the spindle seat **24** of the body **10** formed of brass may be strengthened, so that the sealing between the spindle **40** and the spindle seat **24** may be improved.

[0211] In the foregoing, the disclosure has been shown and described with reference to various embodiments. However, it is understood by those skilled in the art that various changes may be made in form and detail without departing from the scope of the disclosure as defined by the appended claims and equivalents thereof.

Claims

1. A service valve for an air conditioner comprising: a body; a fluid passage configured to penetrate the body; a spindle receiving part configured to communicate with the fluid passage; a spindle screw-connected to the spindle receiving part and configured to open and block the fluid passage; a guard hole configured at a depth on an upper surface of the spindle; and a wrench guard disposed at least partially in the guard hole.
2. The service valve for an air conditioner of claim 1, wherein the wrench guard is formed in a sleeve shape including a through hole, and a cross-section of the through hole is a regular hexagon.
3. The service valve for an air conditioner of claim 2, wherein the guard hole is formed as a polygonal hole, and an outer circumferential surface of the wrench guard is configured as a polygon corresponding to the polygonal hole of the guard hole.
4. The service valve for an air conditioner of claim 3, wherein the polygonal hole is formed as one of: a square hole, a hexagonal hole, and a polygonal star hole.
5. The service valve for an air conditioner of claim 2, wherein the guard hole is configured as an elliptical hole, and an outer circumferential surface of the wrench guard is configured as an elliptical shape corresponding to the elliptical hole of the guard hole.
6. The service valve for an air conditioner of claim 2, wherein the wrench guard includes a stopper provided at an upper end of the wrench guard, and the wrench guard is fixed to the guard hole by at least the stopper.
7. The service valve for an air conditioner of claim 2, wherein the wrench guard includes a fixing protrusion provided on an outer circumferential surface of the wrench guard, and the wrench guard is fixed to the guard hole by at least the fixing protrusion.
8. The service valve for an air conditioner of claim 2, further comprising: a wrench hole formed

below the guard hole.

9. The service valve for an air conditioner of claim 8, wherein a length of the wrench guard is equal to or greater than a depth of the wrench hole.

10. The service valve for an air conditioner of claim 9, wherein the length of the wrench guard is at least 5 mm.

11. The service valve for an air conditioner of claim 1, wherein a material of the body comprises brass, and a material of the wrench guard comprises stainless steel.

12. The service valve for an air conditioner of claim 11, wherein a length of the wrench guard is at least $\frac{1}{2}$ of a depth of a brass wrench hole.

13. The service valve for an air conditioner of claim 1, wherein the body is configured in a cross shape, the fluid passage is on an inside of a leg, a center portion, and a right arm of the body, and the spindle receiving part is configured to connect an upper surface of a head of the body and the center portion.

14. The service valve for an air conditioner of claim 13, wherein the spindle receiving part includes a female screw portion, and the spindle includes a screw portion fastened to the female screw portion.

15. The service valve for an air conditioner of claim 11, wherein a thickness of the wrench guard is from 0.3 mm to 0.5 mm.
